FROM CORRELATION TO CAUSATION

Causal Generative Modeling

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COURSE MODULE

Where this lecture is going

1. Why causal?
2. What makes representations causal?
3. How are representations made causal?
4. How do we model world dynamics causally?
5. How do these causal world models help agents?

Prompt = a cow.



Prompt = a cow on a beach.



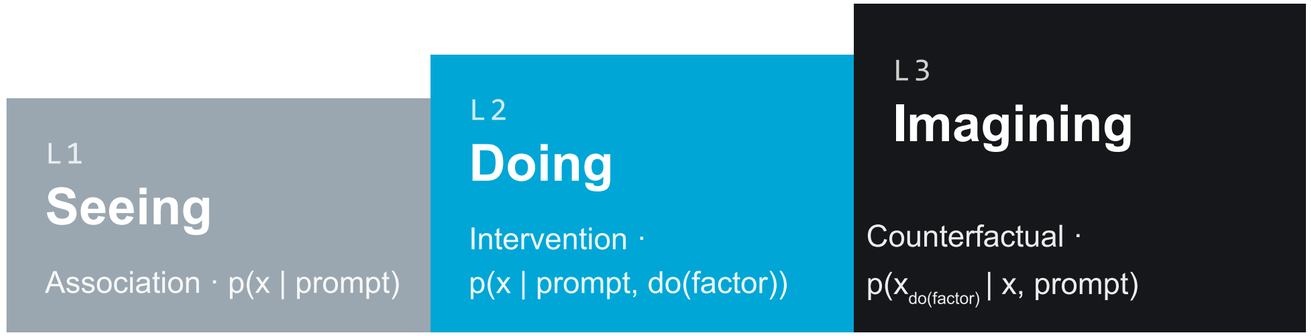
Generative models learn correlations, not causes

VAEs, flows, GANs and diffusion fit a data density `p(x)` and sample from it with remarkable fidelity — by capturing **correlations** among features.

- **Spurious correlations** & shortcut learning — keying on the background, not the object.
  - **Poor out-of-distribution** behaviour when the world shifts.
- **No explanation** of *what* generated the data.

To fix this we climb from *what goes together* to **what produces what**.

Pearl's ladder of causation



Seeing	When I prompt 'a cow' and sample, what images come out?
Doing	If I force the background to be a beach and generate, what cow images come out?
Imagining	This specific cow I just generated on grass, what would it have looked like on a beach?

Each rung needs the one above it: $L1 \subset L2 \subset L3$.

Pearl, 2019 · Bareinboim et al., 2022

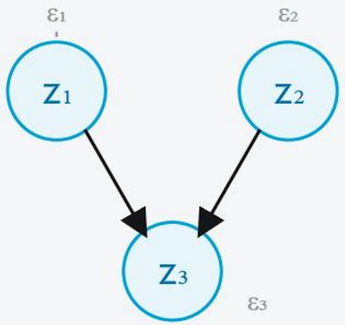
Structural causal models

An SCM generates each variable from its parents and its own noise term:

MECHANISMS & ENTAILED DISTRIBUTION

$$z_i = f_i\left(z_{\mathrm{pa}_i}, \epsilon_i\right) \qquad p(z) = \prod_{i=1}^{n} p\left(z_i \mid z_{\mathrm{pa}_i}\right)$$

ε *The **noise (exogenous) terms** ε_i are everything outside the model that influences z_i — the unmodeled, individual-specific sources of variation.*



$z_3 = f_3(z_1, z_2, \varepsilon_3)$ — an SCM *is* a generative model of the world

Interventions and the do-operator

An intervention *edits a mechanism,* and we re-read the entailed distribution.

Take $z_1 \to z_2$, observationally $p(z_1, z_2) = p(z_1).p(z_2 \mid z_1)$.

HARD INTERVENTION

$$z_2 := c$$

Cuts every parent edge — z_2 is forced to a constant.

$$p(z \mid do(z_2{=}c)) = p(z_1)\,\delta(z_2{-}c)$$

SOFT INTERVENTION

$$z_2 := \tilde{f}_2(z_1, \tilde{\epsilon}_2)$$

Changes the mechanism but keeps the dependence on z_1.

$$p\left(z \mid do(\tilde{f}_2)\right) = p(z_1)\,\tilde{p}(z_2 \mid z_1)$$

IN GENERAL — SURGERY ON THE FACTORIZATION

$$p(z \mid do(z_j{=}c)) = \delta(z_j{-}c)\prod_{i \neq j} p\left(z_i \mid z_{\mathrm{pa}_i}\right) \qquad \text{(the } p(z_j \mid z_{\mathrm{pa}_j}) \text{ factor is removed)}$$

Counterfactuals

A **unit-level** query relative to a factual observation. We observed image x_1 = g(prompt = “a cow”)

- a specific cow on grass and ask: **what would this same cow look like on a beach?**

STEP 1 · ABDUCTION

Infer the noise that explains *this* unit from what we saw.

$$\epsilon_{x_1} \sim p(\epsilon \mid x = x_1,\ \text{prompt})$$

→

STEP 2 · ACTION

Apply the intervention to the SCM.

$$do(\text{factor}_{\text{background}} = \text{beach})$$

→

STEP 3 · PREDICTION

Recompute, reusing the same ε_{x1}.

$$x_2 = g(\epsilon_{x_1},\ do(\text{factor} = \text{beach}))$$

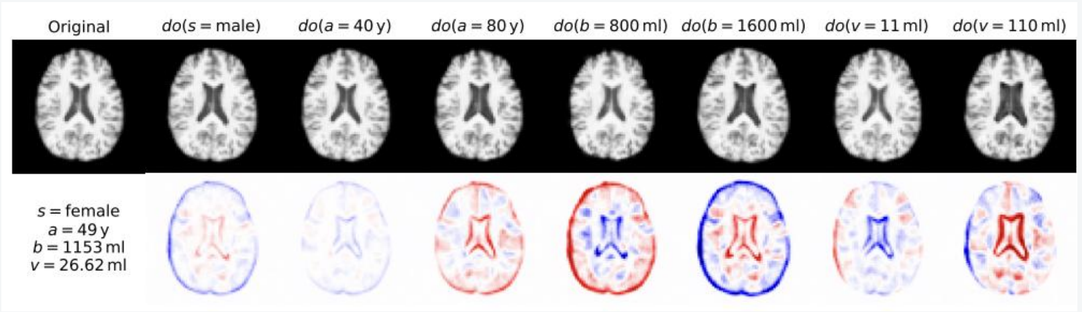
THE COUNTERFACTUAL QUERY

$$p(x_{do(\text{factor}=\text{beach})} \mid x = x_1,\ \text{prompt})$$

Why causality for generative models?

Non-causal controllable generation samples
`p(x | a)` — limited to attribute combinations
seen in training .
A causal model goes further:

- Generate **unseen combinations** , with downstream effects propagated automatically.
- **Edit one attribute** while holding the rest fixed.



Pawlowski et al. (DeepSCM), 2020

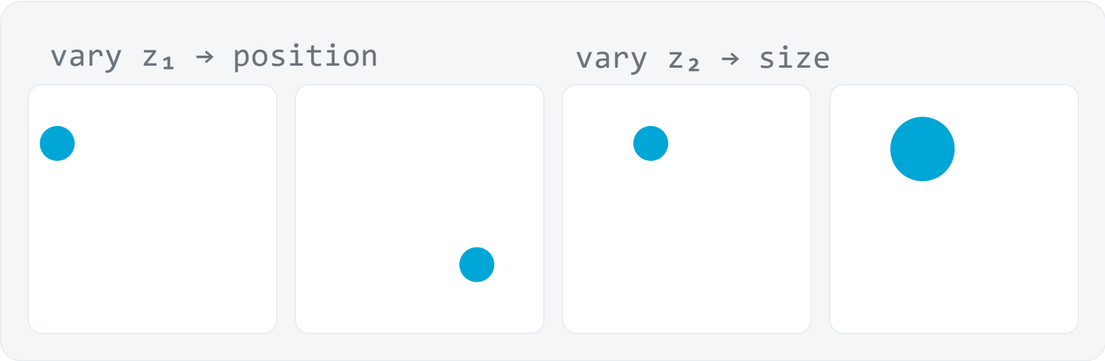
Counterfactual MRI: “Generate the MRI of this brain, had the patient been 40 years older"

Schölkopf et al., 2021

What disentanglement is, and why we need it

A few latent factors generate x through a mixing function: $x = g(z)$.

A representation is **disentangled** when each factor is **axis-aligned** with one coordinate.



Isolated factors are what let us *climb the ladder* inside the latent space:

L1 · See	interpretable, robust associations that don't hinge on spurious features.
L2 · Do	intervene on *one* factor ($do(z_i)$) without disturbing the others.
L3 · Imagine	counterfactuals that imagine a change in a single semantic factor, holding the rest fixed to what was observed.

The identifiability problem

- A standard generative model only ever matches the **marginal** `p(x)`.
- Independence penalties push the codes apart but cannot pin down *which* factors they are.
  - recall KL-divergence with isotropic gaussian in VAE

IDENTIFIABILITY

$$p_\theta(x) = p_{\tilde{\theta}}(x) \implies \theta \sim \tilde{\theta}$$

It guarantees that however often we re-train, we recover the **same factors** up to harmless ambiguities.

A model is **unidentifiable** when different internals explain the same data equally well.

Identifiability theory starts with ICA

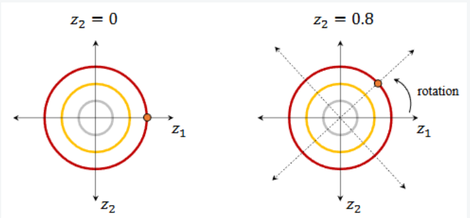
INDEPENDENT COMPONENT ANALYSIS

$$x = g(s), \qquad p(s) = \prod_{i=1}^{n} p(s_i)$$

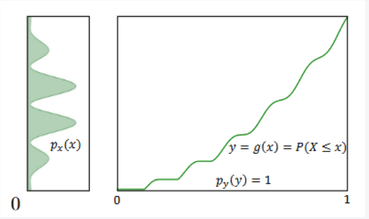
Linear g (`x = As`) is identifiable — provided at most only one source is Gaussian.

Nonlinear g is **not** identifiable: infinitely many independent `s̃ = h(x)` fit equally well.

- **Rotated Gaussian** — invariant to rotation, so every rotation is an equivalent solution. (why PCA's axes aren't identifiable)



- **Darmois construction** — a CDF transform yields independent uniforms, but unrelated to the true sources.



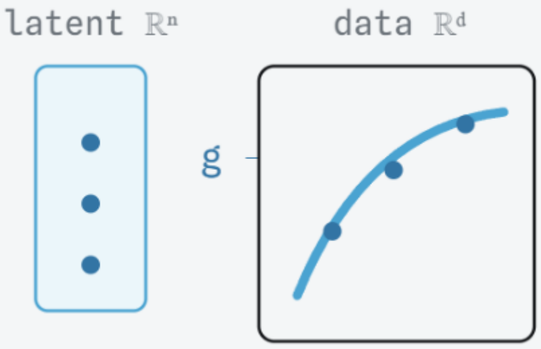
⟹ Unsupervised disentanglement is **impossible** without an inductive bias or auxiliary signal.

Comon, 1994 · Hyvärinen & Pajunen, 1999 · Locatello et al., 2019

Injectivity and the mixing function

We assume the mixing $g : \mathbb{R}^n \to \mathbb{R}^d$ is **injective**, often **diffeomorphic**:

- **Injective** → distinct latents never collapse to the same x; g is invertible *on its image*.
- **Diffeomorphic** → smooth map with a smooth inverse.
- So, inverting to z is **well-posed -** no information is destroyed. Necessary, but not sufficient.

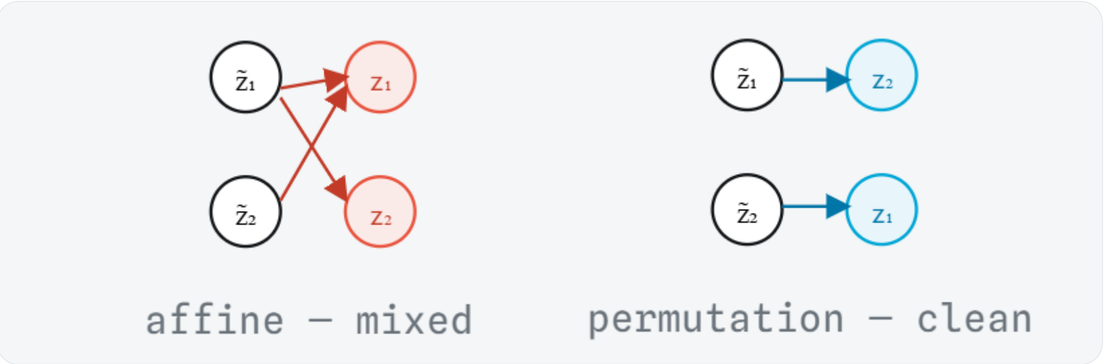


a low-dimensional manifold embedded in high-dimensional space

Identifiable — up to what?

We never recover factors *exactly* , only up to **tolerable ambiguities**. Two milestones:

- **Affine equivalence** — up to an invertible linear map.
  Too weak: still entangled.
- **Permutation equivalence** — reordering + per-axis rescaling.
  This is disentanglement.



Latent conditioning — iVAE

Condition the latents on an observed **auxiliary variable** u (class, domain, time index).
With a conditionally factorized **exponential-family** prior the factors become identifiable.

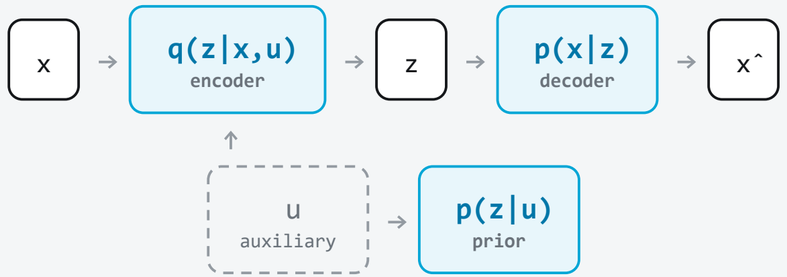
CONDITIONAL PRIOR

$$p(z \mid u) = \prod_i p_\theta(z_i|u)$$

OBJECTIVE — CONDITIONAL ELBO

$$\mathcal{L} = \mathbb{E}_q\big[\log p(x \mid z)\big] - D_{\mathrm{KL}}\big(q(z \mid x, u) \,\|\, p(z \mid u)\big)$$

architecture



RESULT Identifiable up to permutation, given enough variability in u.

Multi-view & contrastive learning

Use **pairs of views** that share a subset S of factors — e.g. a scene before and after a sparse change. The shared structure supplies the variability needed to identify factors.

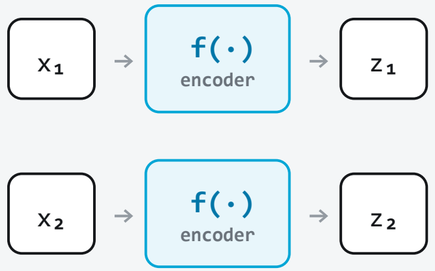
PAIRED GENERATIVE PROCESS

$$z \sim p(z), \quad x_1 = g(z), \quad x_2 = g\big(f(z, \tilde{z}, S)\big)$$

where f keeps z on the shared set S and resamples the rest.

A **contrastive** objective then provably inverts the data-generating process.

two views, shared content



align shared factors · contrast the rest

RESULT Block-identifies the shared *content* latents.

Restricting the mixing — IMA

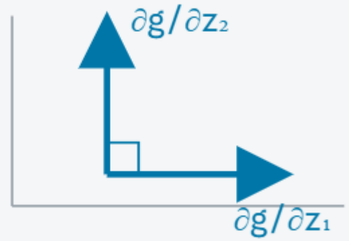
Independent Mechanism Analysis constrains the *geometry* of the decoder: the columns of the mixing Jacobian must be **orthogonal** , so each source influences the output through its own channel.

IMA CONDITION

$$\log |J_g(z)| = \sum_{i=1}^{n} \log \left\| \frac{\partial g}{\partial z_i}(z) \right\|$$

Added as a regularizer on the decoder Jacobian, it shrinks the solution space toward the true factors — without any labels.

orthogonal Jacobian columns



RESULT An inductive bias that yields identifiability with no auxiliary data.

Sparsity

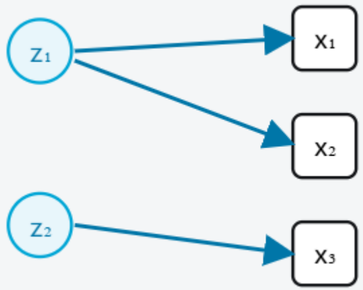
Structural sparsity: for each latent z_i there is a set of observed variables that *only it* generates. Sparse wiring between sources and observations breaks the symmetry.

SPARSE MIXING SUPPORT

$$\forall i,\ \exists x_k : \frac{\partial x_k}{\partial z_j} = 0\ \forall j \neq i$$

Variants add sparse *mechanisms* between latents, or a sparse temporal action graph.

sparse mixing — z → x



x₃ is generated by z₂ alone

RESULT Disentanglement from sparsity — no labels, no pairs.

The causal ingredients

Causal representation learning keeps identifiability:

FACTORS MAY BE CAUSALLY DEPENDENT

$$p(z) = \prod_{i=1}^{n} p(z_i \mid z_{\mathrm{pa}_i})$$

- **Independent Causal Mechanisms** — each mechanism is autonomous; intervening on one leaves the others unchanged.
- **Sparse Mechanism Shift** — distribution shifts touch only a few mechanisms → built-in robustness.

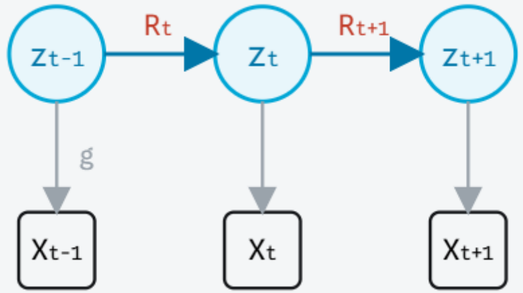
From static factors to dynamics over time

In an interactive environment we observe a **sequence** of images X_t and the agent's **action** R_t. Hidden causal variables z_t evolve like a dynamic Bayesian network and generate each frame, $X_t = g(z_t)$.

A LEARNED, ACTION-CONDITIONED TRANSITION MODEL

$$p(\{X\} \mid \{R\}) = \int p(z_0) \prod_{t=1}^{T} \underbrace{p_\omega(z_t \mid z_{t-1}, R_t)}_{\text{transition}} p_g(X_t \mid z_t)\, dz$$

The same identifiability question returns — now *across time, conditioned on actions*.



latent states transition under actions; each emits an observation

BISCUIT — identifiability from binary interactions

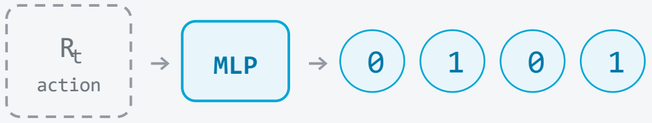
- **Key assumption**: the agent's effect on each causal variable is a **binary interaction** $I_t^i \in \{0,1\}$
- This means, every variable has *two* mechanisms (observational vs. interventional), like a soft intervention.

INTERACTION = UNKNOWN FUNCTION OF THE ACTION

$$I_t^i = f_i(R_t, C^{t-1}) \in \{0,1\}$$

e.g. pressing a button, opening a door, colliding with an object — each *switches* a variable's dynamics.

action → binary interaction mask → mechanism per latent



Distinct interaction patterns (+ variability) make causal variables identifiable up to permutation — breaking the Gaussian rotation symmetry.

BISCUIT — a causal transition prior

A VAE with a **structured prior** : per latent, an MLP infers the interaction $\hat{I}_t^i$ from (z_{t-1}, R_t) , then a second MLP outputs that latent's next-step distribution.

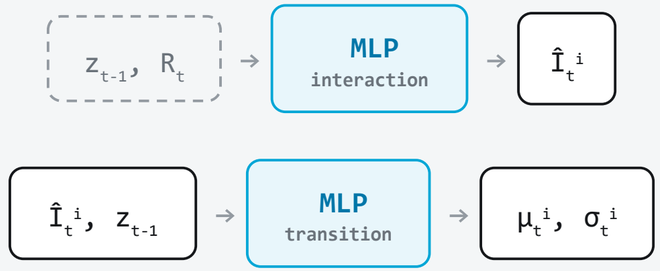
FACTORIZED, ACTION-CONDITIONED TRANSITION PRIOR

$$p_\omega(z_t \mid z_{t-1}, R_t) = \prod_i p(z_t^i \mid z_{t-1}, \hat{I}_t^i)$$

OBJECTIVE — ELBO OVER OBSERVATION TRIPLETS

$$\mathcal{L}_t = -\mathbb{E}_{q_\phi}\big[\log p_\theta(x_t \mid z_t)\big] + \mathbb{E}\left[D_{\mathrm{KL}}\big(q_\phi(z_t \mid x_t) \,\|\, p_\omega(z_t \mid z_{t-1}, R_t)\big)\right]$$

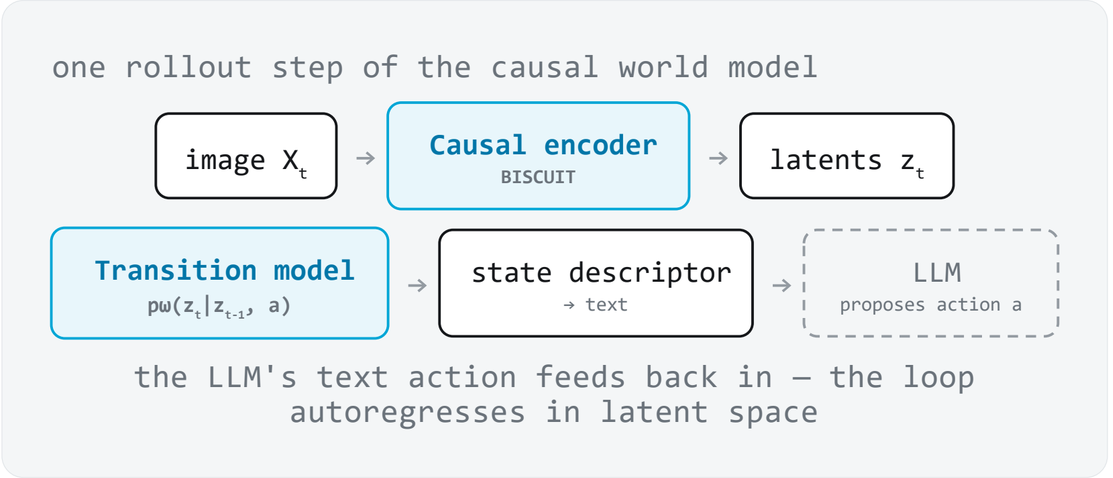
prior structure (per latent i)



each latent = a causal variable the action can switch → controllable generation of unseen states

Language agents meet causality

- An LLM's commonsense causality is often incomplete or wrong for a *specific* environment.
- So, wrap a BISCUIT **causal world model** around it acting as a learned **simulator** the LLM can query.
- Causal latents are **grounded in natural language** — a flexible interface for the LLM.
- **Text-based actions** replace hand-coded action encodings — intuitive and data-efficient.



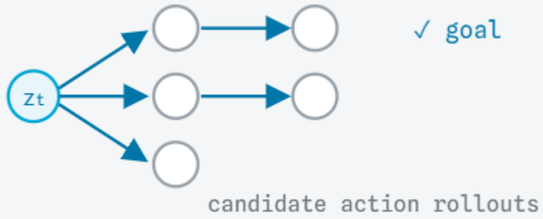
Planning with action-conditioned causal rollouts

THE WORLD MODEL, WITH TEXT ACTIONS L_T

$$p(\{E\} \mid \{L\}) = \int p(z_0) \prod_t p_\omega(z_t \mid z_{t-1}, L_\epsilon(L_t))\, p_\phi(E_t \mid z_t)\, dz$$

- The LLM **rolls out candidate actions** in latent space and evaluates several possible futures before committing.
- BISCUIT's identifiability makes those predicted effects **reliable**, not hallucinated.

RESULT Causally-aware planning **outperforms LLM-only reasoners** — and the gap *grows with the planning horizon* .

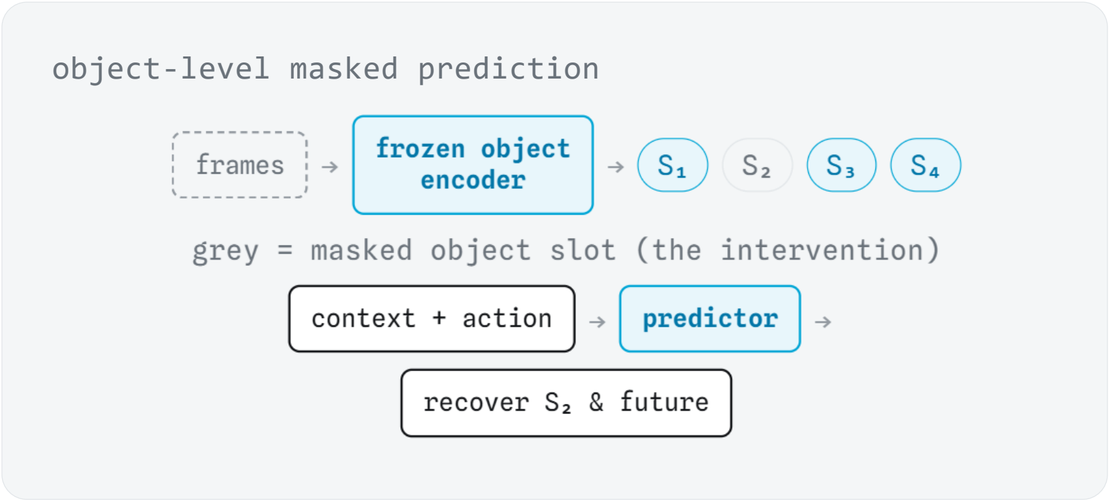


Causal-JEPA — world models by object-level masking

JEPA predicts world states in *representation space* , never pixels. C-JEPA runs it over **object-centric** latents and enforces causal learning by **masking object-level latents** as a latent intervention.

- Each masked object must be inferred from the others → **counterfactual-like** prediction queries.
- This blocks shortcut solutions (trivial interpolation) and makes **interaction reasoning necessary**.

Assumes an object-centric encoder; actions enter as optional auxiliary variables.



That’s all folks